

Canvas Protocol — Product Spec

Version: 0.1 (MVP) **Authors:** Rohit Sarkar, Mateo **Status:** Pre-build · For internal alignment

Overview

Canvas Protocol is a decentralized verification marketplace that replaces standard Telegram group join checks with meaningful captcha tasks. Every time a human tries to join a registered group, they complete a short task sponsored by an advertiser or AI lab. The task is matched to the group's topic. Completion is validated by Kimi inference. Payment is released automatically to the group owner via smart contract on Base.

The result is a two-sided market: advertisers get verified, contextual human completions; group owners get paid to grow their communities; humans get access without meaningless puzzles.

Problem

Standard Telegram verification wastes human attention. Users solve arbitrary bot-checks that generate no value for anyone. Group owners earn nothing. Advertisers have no way to reach crypto-native audiences inside messaging. AI labs need human-verified training data but existing pipelines (Scale AI, Mechanical Turk) lack audience context and are expensive to operate. The human attention used in group verification is real, recurring, and currently unmonetized.

Solution

Replace Telegram join verification with a sponsored captcha task. The task serves as both the verification mechanism and the value-generating interaction.

For advertisers and AI labs: a pay-per-verified-completion channel reaching humans with known context and interests. A DeFi user ranking protocol responses is more valuable training data than a random Mechanical Turk worker doing the same task.

For group owners: recurring revenue from verification activity, with no change to the user experience beyond the task replacing a standard bot-check.

For humans: the same frictionless entry they expect, but with a task that takes 10-15 seconds and is relevant to the group they're joining.

Stakeholders

Advertiser / AI lab — buys a batch of verifications (e.g. 100 at \$0.10–\$0.25 each) via a natural language conversation with the Canvas Bankr agent. Funds are locked in a smart contract escrow. Budget draws down automatically as completions occur. Receives completion data and response signals after each verification.

Group owner — registers their Telegram group and payout wallet via the Canvas Bankr agent. Adds the Canvas bot as an admin. Earns USDC per verified completion. Re-verification events (periodic re-checks of existing members) create ongoing recurring revenue.

Human — attempts to join a Telegram group. Is muted by the bot and receives a captcha task in a private DM. Completes the task. Is admitted if Kimi scores the response as genuine.

Canvas Bankr agent — the natural language interface for both sides of the marketplace. Handles advertiser buy flow, group owner registration flow, and ongoing reporting.

Telegram bot (@canvasbot) — intercepts join requests, mutes users, serves captcha tasks in private DMs, receives responses, and handles admission or re-prompt. Must be added as admin to any registered group.

Agent server (Railway) — orchestrates the full per-verification flow. Hosts the Telegram bot logic, Postgres database, Kimi inference calls, and contract call triggers. Always-on service.

Smart contract (Base) — non-custodial escrow. Holds advertiser USDC. Logs each completion attempt onchain (step 1). Releases payout to the group owner wallet on a valid Kimi score (step 2).

How It Works

Setup (one-time per party)

Group owner onboarding:

1. Group owner DMs Canvas agent on Bankr
2. Agent asks for: group link, group topic/interests, payout wallet address, price per verification (default \$0.10)
3. Agent writes group to the Postgres registry
4. Group owner adds @canvasbot as admin to their Telegram group
5. Group is live and eligible to receive advertiser budgets

Advertiser onboarding:

1. Advertiser or AI lab DMs Canvas agent on Bankr

2. Agent shows available groups filtered by topic and member count
 3. Advertiser selects groups, specifies budget (e.g. 100 verifications)
 4. Advertiser funds escrow via Bankr — USDC locked in smart contract
 5. Budget is earmarked for the selected groups
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Per-Verification Flow

1. Human attempts to join a registered Telegram group
 2. Telegram webhook fires to the agent server
 3. Bot checks Postgres: confirms group is registered and has active advertiser budget
 4. Bot mutes the human and sends a captcha task in a private DM, matched to the group's topic and advertiser category
 5. Human completes the task and replies
 6. Agent server fires **step 1**: completion logged onchain on Base (attempt is recorded regardless of quality)
 7. Response is sent to Kimi inference for scoring
 8. If Kimi scores the response as genuine: agent server fires **step 2** — USDC released from escrow to group owner wallet
 9. If Kimi fails the response: human is re-prompted once, then denied entry if still invalid. No payout
 10. Human is unmuted and admitted to the group
 11. Advertiser is notified via Canvas agent: completion count, response data, remaining budget
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Re-Verification (Phase 2)

Existing group members are periodically re-verified (e.g. every 30 days). Each successful re-verification triggers a payout to the group owner. This creates ongoing recurring revenue tied to community size and retention, not just new joins. Mods and admins can be set up to receive a share of re-verification payouts.

Captcha Task Examples

Type 1 — RLHF training data (AI lab advertiser)

Group context: DeFi / crypto trading

Task:

Rank these three responses from most to least helpful. Reply with the order (e.g. B, A, C).
A — "To swap tokens, connect your wallet, select the pair, and confirm the transaction." **B** — "You need ETH for gas. Approve the token first, then swap. Check slippage before confirming." **C** — "Blockchain technology enables decentralised finance through trustless smart contracts."

Why it works: A crypto-native human immediately recognises B as the most practically useful. A bot randomly clicking doesn't. The label is high-quality because the annotator has verifiable context. AI labs pay a premium for this over generic annotation pipelines.

Type 2 — Agent acquisition (DeFi protocol advertiser)

Group context: Base DeFi traders

Task:

You have 10,000 USDC to allocate. What would you do?
A — Supply to Aave at 4.2% APY and borrow against it **B** — Hold in a cold wallet and wait for better rates **C** — Bridge to another chain for a higher yield farm
Sponsored by Aave · no wrong answer

Why it works: There is no wrong answer — the advertiser doesn't need correct responses, they need verified DeFi-native humans to engage with their brand and product. After verification, the bot offers the Aave agent connection. Advertiser pays per verified lead, not per ad impression.

Technical Architecture

Stack

Layer	Technology
Agent server	Node.js / TypeScript · Railway
Telegram integration	Telegram Bot API · @BotFather token · bot is group admin
Database	Postgres (groups registry, verification queue, budgets, user state)
AI validation	Kimi API (external, pay per call)
Agent interface	Canvas agent on Bankr
Payments	USDC on Base via x402

Layer	Technology
Smart contract	Solidity on Base mainnet

What Mateo Already Built (from Basemate v2 / tg-branch)

- Telegram bot framework with webhook handling
- Intent detection and group matching logic
- Redis + Postgres infrastructure on Railway
- x402 USDC payment integration
- XMTP agent infrastructure (separate, not used here)

What Needs to Be Built Net-New

Canvas Bankr agent

- Buy flow: group selection by topic, budget input, escrow funding trigger
- Register flow: group link intake, wallet collection, Postgres write
- Reporting: completion counts, budget remaining, response data

Bot verification module

- Join intercept and mute logic (join.ts handler — currently a spec, not code)
- Captcha task selection engine (match task type to group topic + advertiser)
- Task serving via Telegram DM
- Response receiver and parser

Kimi inference integration

- HTTP call to Kimi API with task + response
- Score parsing and threshold check
- Pass/fail routing to contract calls or re-prompt

Smart contract

- Escrow deposit function (advertiser funds)
- Step 1: log completion onchain
- Step 2: release payout to group owner wallet (gated on Kimi pass)
- Budget tracking per group per advertiser
- Refund function for unused budgets

Contract call wiring (in agent server)

- Step 1 call triggered on response receipt
 - Step 2 call triggered on Kimi pass
 - RPC client connecting to Base
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MVP Scope

Phase 1 — Prove the loop (no real money)

- Telegram bot intercepts join requests in a test group
- Bot serves a hardcoded captcha task in DM
- Kimi validates the response
- Bot admits or re-prompts
- Contract calls are stubbed (logged to console, no real onchain calls)
- Advertiser is a mock wallet seeded by the team

Goal: end-to-end verification loop working in a live Telegram group

Phase 2 — Real money, real contract

- Smart contract deployed to Base mainnet
- Bankr buy flow wired to real escrow funding
- Contract calls fire on real Base transactions
- Group owner receives real USDC on verification
- Canvas Bankr agent handles registration and buying via natural language

Goal: first real payout to a group owner from a real advertiser

Phase 3 — Marketplace and growth

- Re-verification cadence per group
 - Mod/admin payout splits
 - Multiple task types and advertiser categories
 - Group owner dashboard (or expanded Bankr agent reporting)
 - Outbound to AI labs as first advertiser category
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Business Model

Protocol fee: Canvas takes a percentage of each verification payout (target: 10–15%). This is taken at the smart contract level before the group owner receives their share.

Moat: The defensible position is not the fee but the group registry and quality verification layer. Canvas becomes the trust infrastructure — the thing that tells advertisers which groups are legitimate and which completions are genuine. Every group listed anywhere has to come back to Canvas to be credible.

Long-term: Re-verification creates recurring revenue proportional to total active group membership across the protocol, not just new joins. A group with 5,000 members re-verified monthly at \$0.10 each generates \$500/month passively, of which Canvas keeps \$50–75.

Project Structure

This is a new project, separate from Basemate. New repo, new entity, clean equity split established before any code is written.

Rohit — product design, marketplace spec, advertiser-side business development, content and go-to-market, Bankr ecosystem relationships

Mateo — technical infrastructure lead, Telegram bot, agent server, Railway deployment, smart contract integration

Bankr team — smart contract rails, x402 payment integration, agent SDK

Open Questions

- What is the equity / token split between Rohit and Mateo?
 - Does the smart contract need an external auditor before Phase 2 goes live?
 - What is the minimum group size to be eligible for registration (to prevent sybil groups)?
 - How long is the escrow lock period, and what triggers a refund to the advertiser?
 - What task types does Phase 1 support — ranking only, or also binary choice and free text?
 - Who are the first two or three AI labs to approach as Phase 2 advertisers?
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